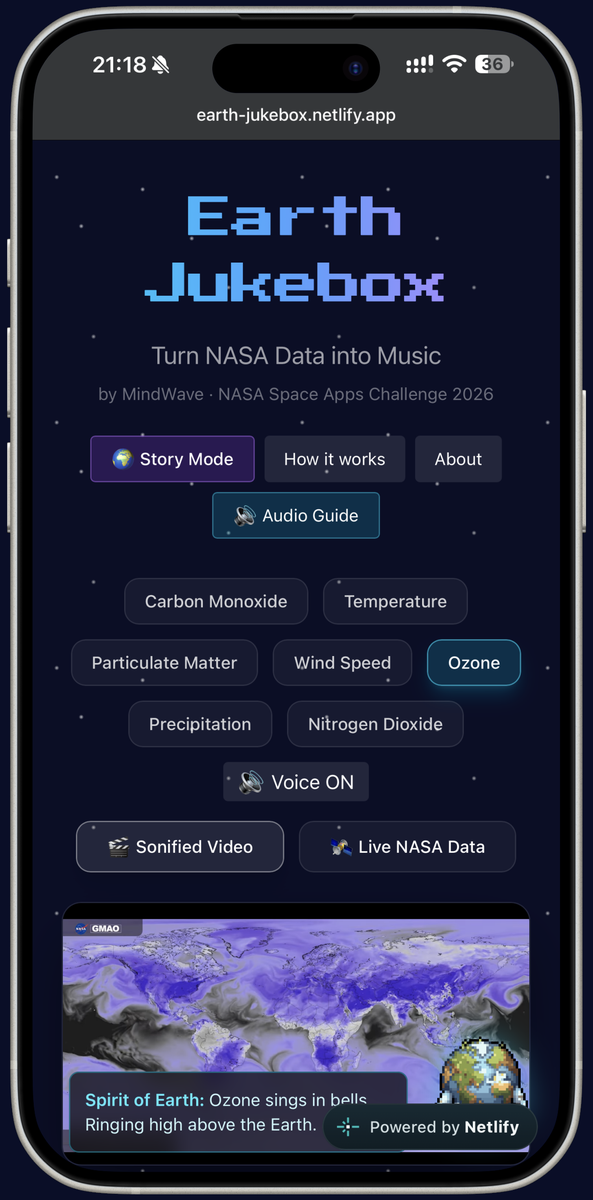


EARTH JUKEBOX

TURNING NASA EARTH OBSERVATION DATA INTO MUSIC

MINDWAVE · NASA SPACE APPS CHALLENGE

→ GITHUB ←

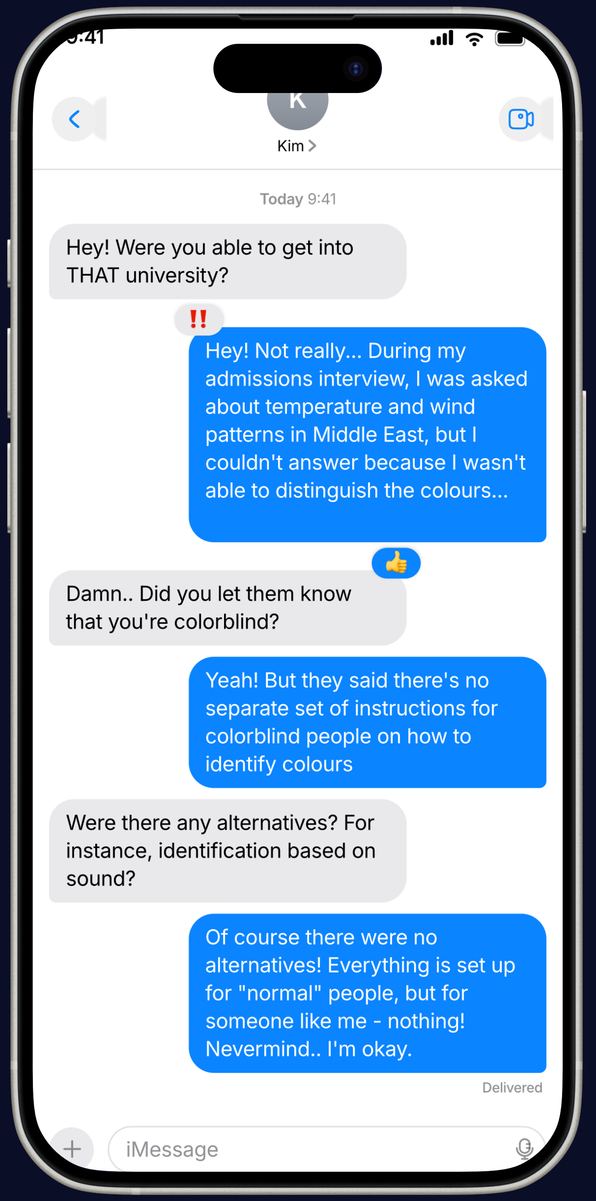
→ MVP ←

PROBLEM & SOLUTION

Problem: Climate Data Beyond Sight

- More than 300 million blind or visually impaired people worldwide are unable to see complex heat maps and graphs from NASA
- People with dyslexia, colorblind people, auditory learners, and children also struggle to process dense visual content

Solution: Transform the colours in satellite imagery into musical instruments.



HOW IT WORKS (CORE IDEA)

Each colour is the Visual Tool

The user hears “heavy bass” -
which means that in the
atmosphere there is a high
concentration of carbon monoxide

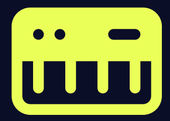
Blue → Cello
Purple → Synth
White → Bells






Red → Bass
Orange → Piano
Yellow → Strings
Green → Flute





ACCESSIBLE BY DESIGN. USEFUL FOR EVERYONE

Blind and visually impaired individuals: sound serves as their primary channel for perception

People with dyslexia: sound bypasses the challenges associated with reading text

Auditory learners: they retain information better when they hear it

Children: music makes abstract scientific concepts concrete and engaging

MORE THAN JUST A VIDEO – AN INTERACTIVE PLATFORM

Features:

- Sonified Videos (seven NASA datasets) 🎵
- Instrument Mixer: isolate the sound of a specific colour 🎛️
- Audio Guide: voice-guided navigation and descriptions 🗣️
- Ask the Earth (AI Chatbot): get answers to your questions from the perspective of the "Spirit of the Earth" (powered by the Gemini API) 🤖
- Guess the Instrument: an educational mini-game 🎮



TECHNOLOGY STACK

FRONTEND



React 18

Vite





Tailwind CSS 4

Framer Motion

Recharts

React-Leaflet

BACKEND



Node.js

Netlify Functions

(serverless)



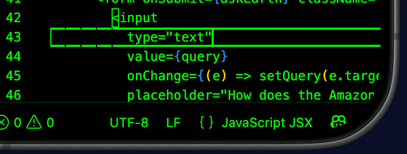
AI & PIPELINE

Google Gemini API (Fallback on 3 models)



Python (OpenCV for colour analysis, music21, FluidSynth, FFmpeg) - offline





5. Sync audio + video (FFmpeg).

Mode 2 — From video (MP4)

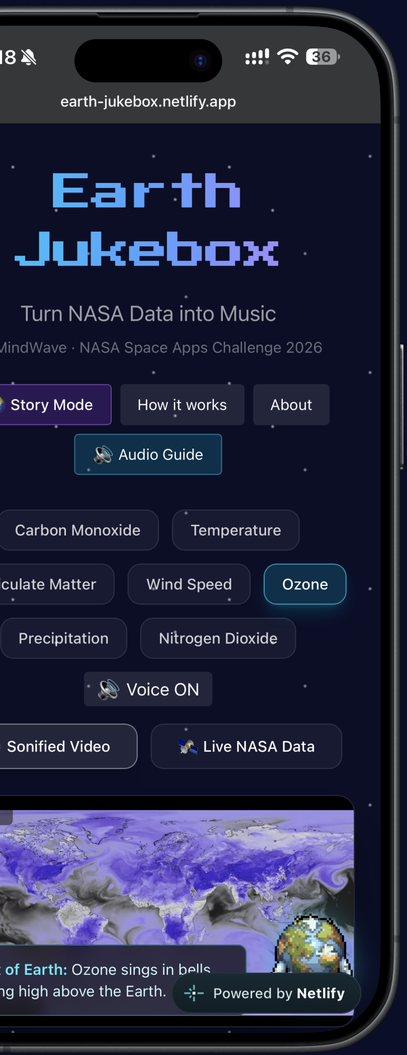
1. Extract frames from video (12 fps).
2. Analyze color content per frame.
3. Generate MIDI with 7 tracks.
4. Render and sync.

Color → Instrument

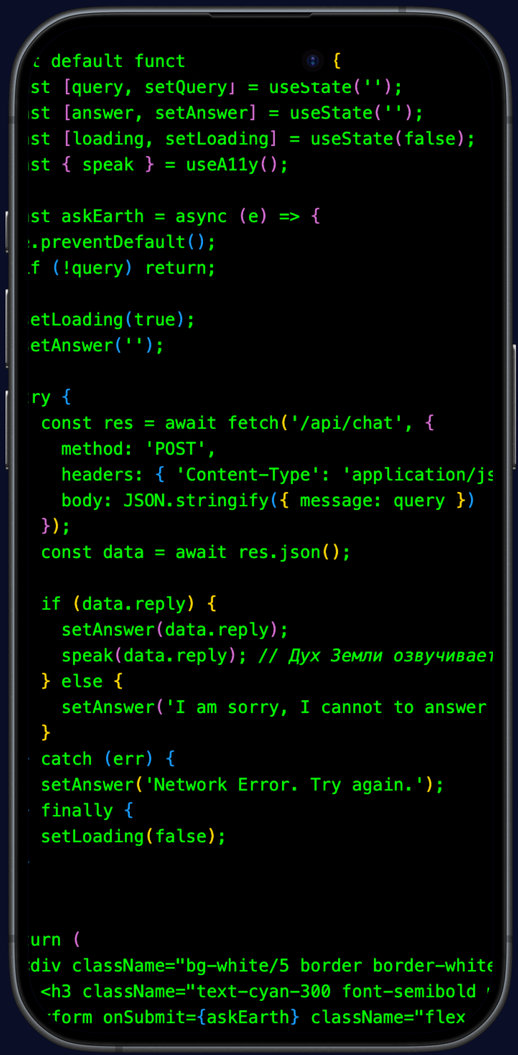
- Red — Bass
- Orange — Piano
- Yellow — Strings
- Green — Flute
- Blue — Cello
- Purple — Synth
- White — Bells

Close

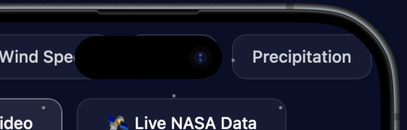
Powered by Netlify

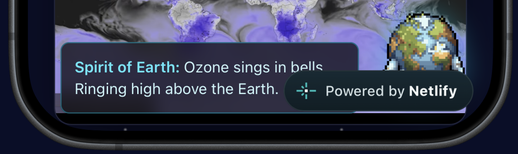








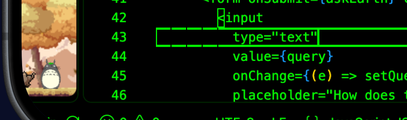


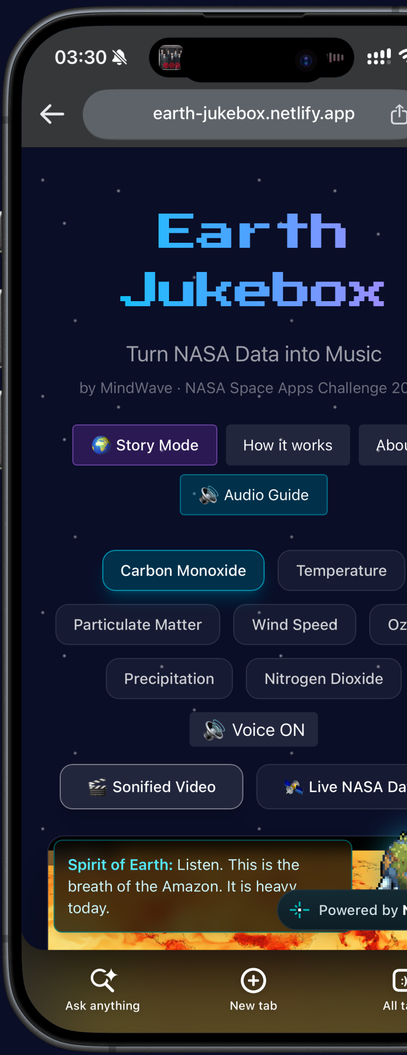


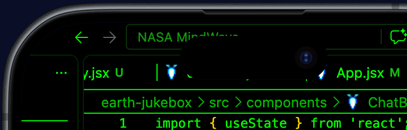












Data Pipeline (How the music is born)

- Take a video or image from gmao.gsfc.nasa.gov - either as a downloaded file or via a source link (API/JSON)

VIDEO

- Perform an offline analysis of the video to precisely determine the proportion of each colour

OPENCV (HSV-ANALYS)

- A script analyses every pixel in each frame and calculate the percentage breakdown of colours

% COLOURS

- Next, a Python script generates a MIDI file, with each colour assigned to its own separate track

MUSIC21

- FluidSynth converts the MIDI data into audio

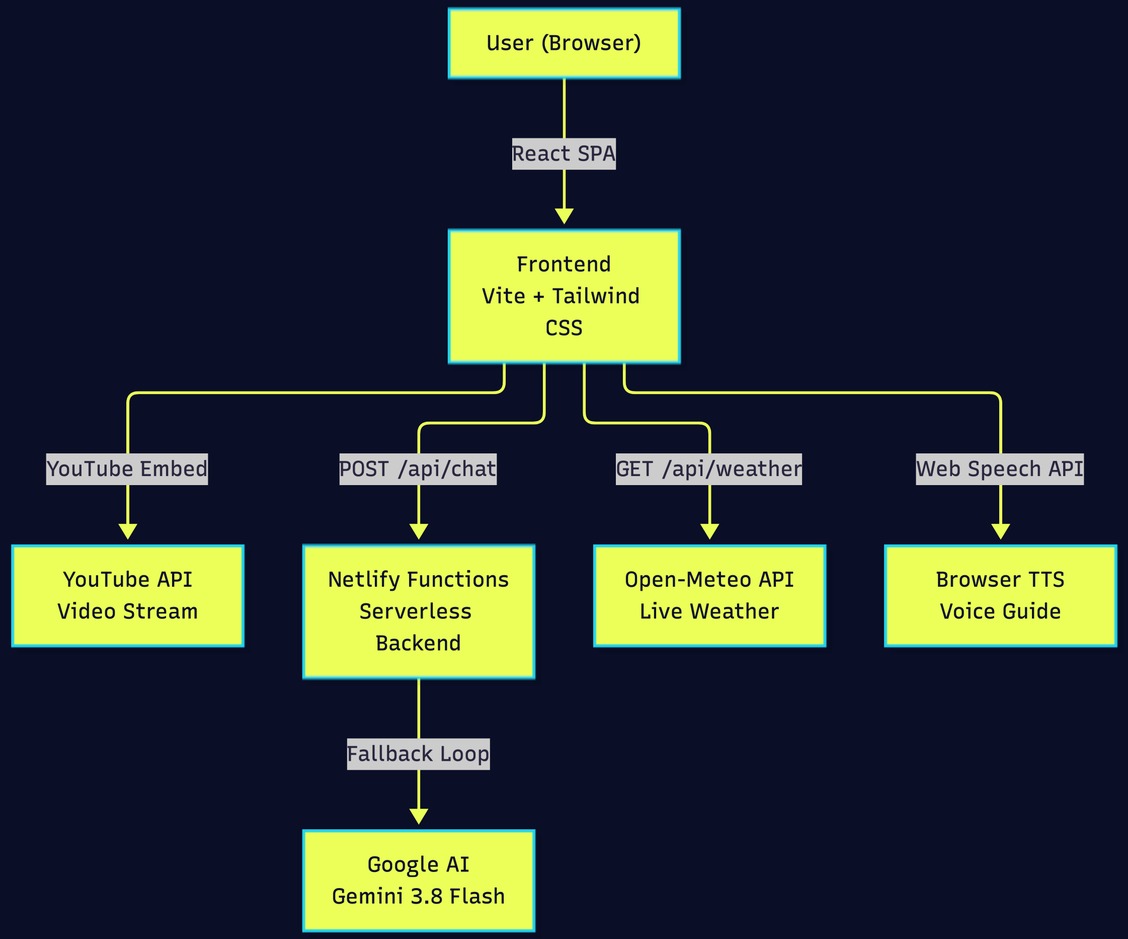
FLUIDSYNTH

- FFmpeg then merges the audio with the video

FFMPEG

SOLUTION ARCHITECTURE.

MVP TRANSPARENCY



- Also includes integration with Open-Meteo and the Web Speech API.
- Real-time color analysis is not possible for YouTube (due to CORS) - this design choice was made to efficiently handle large volumes of data.
- We use static percentages calculated offline (these are accurate, verified data points).
- The instrument mixer currently only works for local video files.
- The NASA GIBS WMS service is occasionally blocked in certain regions.

Note: We are transparent about these limitations in the README

SCALING. 5-YEAR PLAN

Year 1: Integration of all NASA GMAO data: support for tactile (Braille) displays.

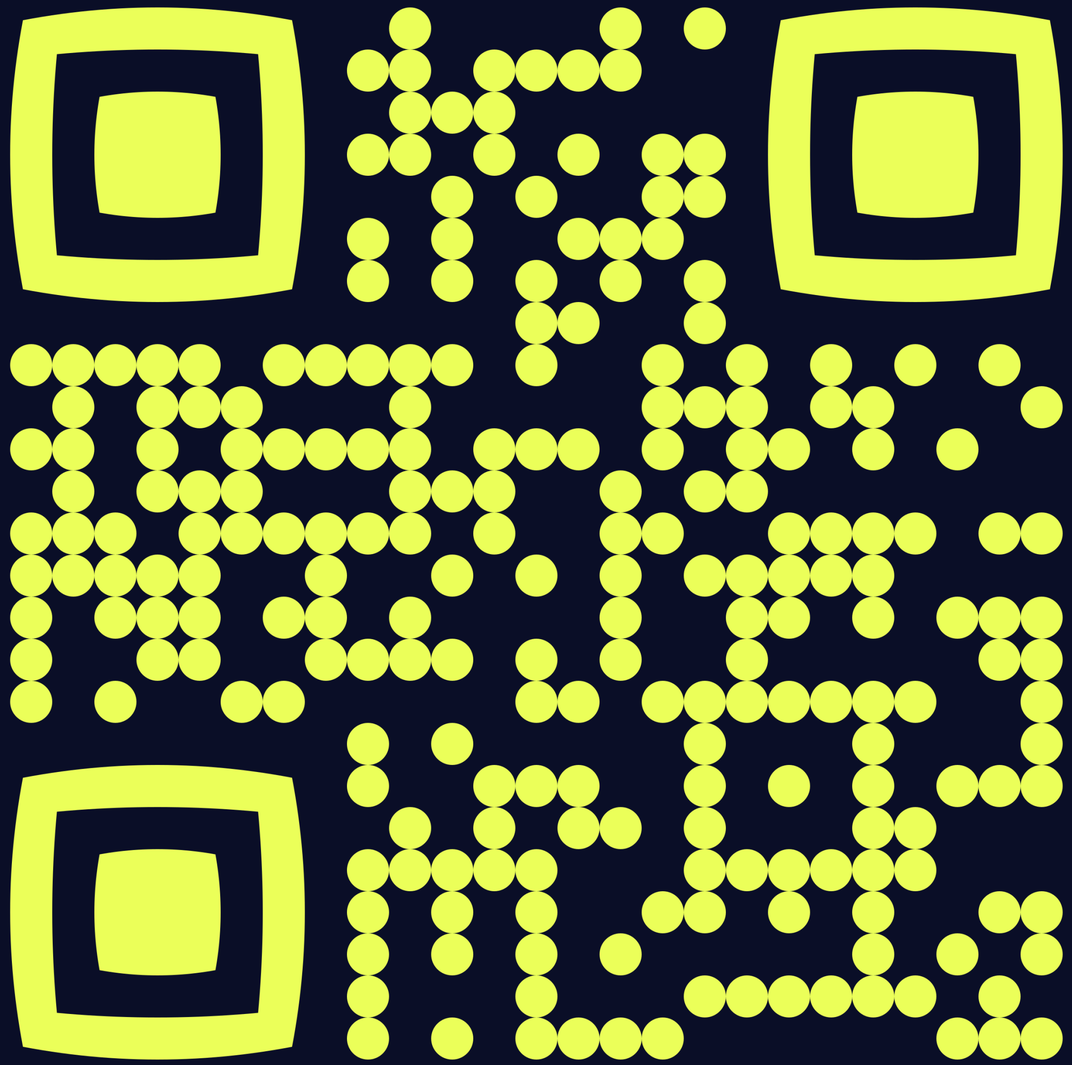
Year 2: Real-time synthesis via the Web Audio API.

Year 3: AI-adaptive soundscapes (using ML models).

Year 4: Educational portal and API for schools.

Year 5: Open standard for scientific sonification.

EARTH JUKEBOX



VERSION: 1.0.0

SITE →

TEAM:
MINDWAVE
WINTER SOFIA-LUNA ⟨FULLSTACK⟩
→ GITHUB ←